# ABEE2007

# Environmental Science for Architects Autumn Semester Project Submission

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|---|---|
| Surname: | Enhos |
| Forename: | Derin |
| Student ID: | 20481498 |
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| | | |
|---|---|---|
| 1 | Do you have an approved EC for this submission? | No |
| 2 | Do you have an approved study support plan for this submission? | No |
| 3 | Do you have an EC pending for this submission? | No |
| 4 | Do you have a study support plan pending for this submission? | No |
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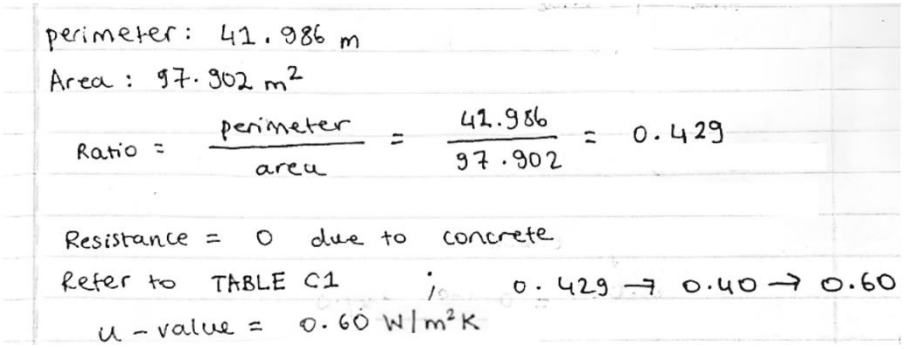
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## Envelope Performance

**U- Value calculations for building materials of my client's house:**

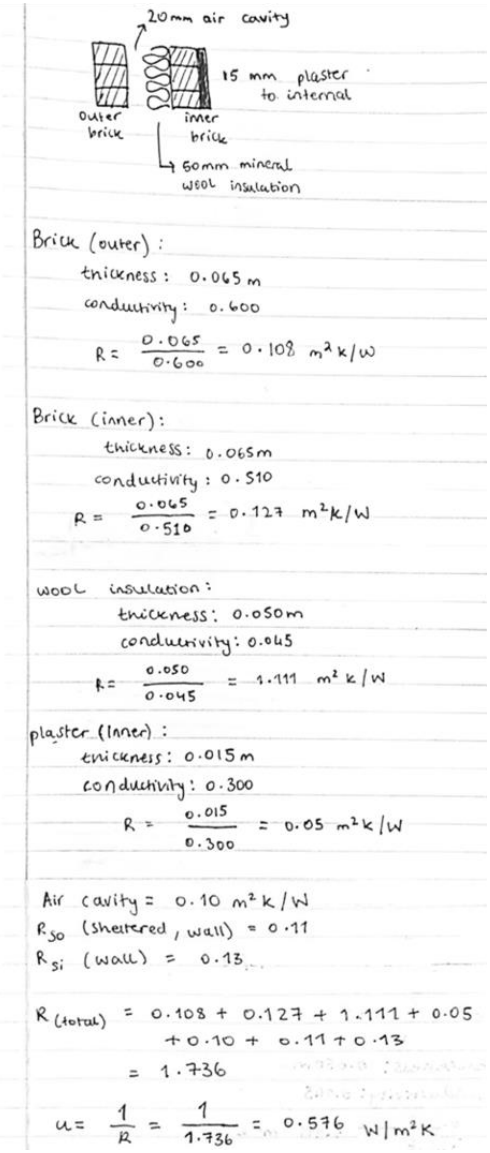
| Materials | Conductivity (W/m.K) | Resistance ($m^2$ .K/W) | U-Value ($W/m^2$ .K) |
|---|---|---|---|
| Brick (outer) | 0.600 | 0.1083 | 9.23 |
| Brick (inner) | 0.510 | 0.1275 | 7.84 |
| Concrete (cast) | 1.520 | 0.0329 | 30.38 |
| Plasterboard | 0.210 | 0.0595 | 16.81 |
| Plaster | 0.300 | 0.0500 | 20.00 |
| Timber (softwood) | 0.150 | 0.3333 | 3.00 |
| Mineral wool quilt insulation | 0.045 | 1.1111 | 0.90 |

## Floor U-value




Approved Document Part L requires a U-value of 0.13 for floors, but the floor has a U-value of 0.60, failing to comply. The concrete floor transfers significant heat to the outside. Key contributors to the high U-value include:

1. The lack of insulation, resulting in zero resistance and allowing heat to flow out easily.
2. The floor's construction, consisting of 50mm concrete screed on 100mm oversite concrete. Concrete, a thermal mass, absorbs and stores heat, but its impact on increasing the U-value is minimal.

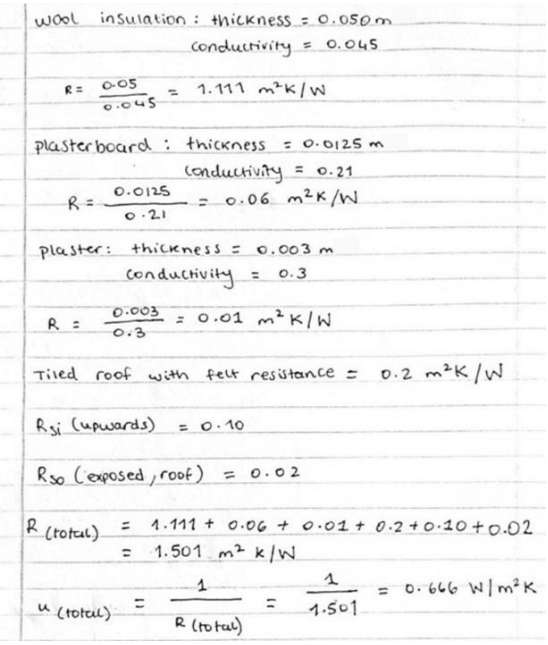



## Wall U-value

Approved Document Part L requires a wall U-value of 0.18, but the wall's U-value is 0.576, failing to comply. The following factors contribute to this higher U-value:

- Outer and inner brick layers have low thermal conductivity.
- An unventilated air cavity adds thermal resistance by trapping air and reducing heat transfer.
- Wool insulation has low thermal conductivity, but its 50mm thickness is insufficient, raising the U-value. Increasing it to 100mm can reduce the U-value to 0.34, closer to Part L.
- Plaster, though thin, adds minimal resistance due to its low thermal conductivity.

Thicker or higher-density insulation, like polystyrene foam, can further reduce heat flow and lower the U-value.

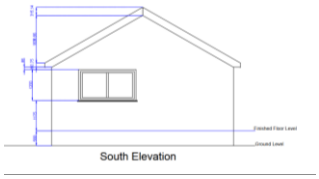



## Roof U-value:

- Part L requires a roof U-value of 0.11, but the roof's U-value is 0.666, failing to comply. The insulated ceiling gives the pitched roof the highest U-value compared to the floor and walls. Warm air rises to the roof, and its construction contributes to the high U-value.

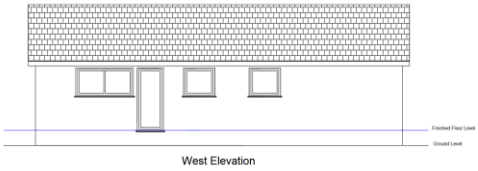
- The high U-value is mainly due to insufficient wool insulation (50mm) and its placement between joists, causing thermal bridging and easy heat escape.
- Limited thickness of plaster, plasterboard, and insulation reduces overall thermal resistance, increasing heat loss and raising the U-value.

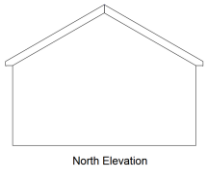
## HLC values for building elements



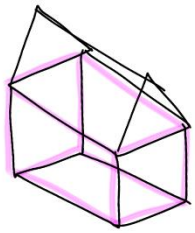







**South elevation**




**Total elevation area** = 2.45 x 6.993 = 17.133 $m^2$

**Area of openings** = Window (1) 1.2 x 2.2 = 2.64 $m^2$

**Area of wall** = 17.133 – 2.64 = 14.493 $m^2$

**East elevation**

**Total elevation area** = 2.45 x 14 = 34.3 $m^2$

**Area of openings** = Window (2) 2.118 x 2.2 = 4.66 $m^2$, Door (1) 2.375 x 1 = 2.375 $m^2$

Window (3) 1.075 x 0.5 = 0.538 $m^2$, Window (4) 2.118 x 2.2 = 4.66 $m^2$

Total = 4.66 + 2.375 + 0.538 + 4.66 = 12.233 $m^2$

**Area of wall** = 34.3 – 12.233 = 22.067 $m^2$

**West elevation**

**Total elevation area** = 2.45 x 14 = 34.3 $m^2$

**Area of openings** = Window (5) 1.075 x 2.2 = 2.365 $m^2$, Door (2) 2.375 x 1 = 2.375 $m^2$

Window (6) 1.075 x 1.2 = 1.29 $m^2$, Window (7) 1.075 x 1.2 = 1.29 $m^2$

Total = 2.365 + 2.375 + 1.29 + 1.29 = 7.32 $m^2$

**Area of wall** = 34.3 – 7.32 = 26.98 $m^2$

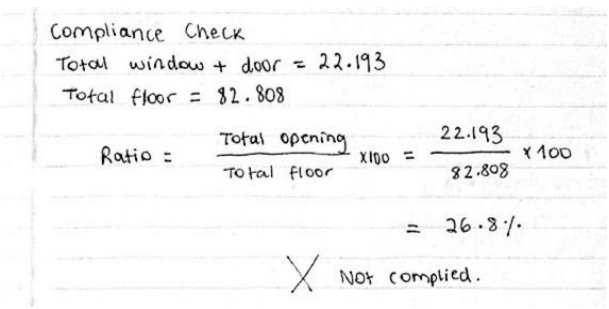
**North elevation**

**Total elevation area** = 2.45 x 6.993 = 17.133 $m^2$

**Area of openings** = 0 $m^2$

**Area of wall** = 17.133 $m^2$

***Total area of walls = 14.493 + 22.067 + 26.98 + 17.133 = 80.673 $m^2$***


Compliance Check
Total window + door = 22.193
Total floor = 82.808
Ratio = $\frac{\text{Total opening}}{\text{Total floor}} \times 100 = \frac{22.193}{82.808} \times 100$
= 26.8%
✗ Not complied.

Part L indicates that opening areas for a dwelling shouldn't exceed a total area of openings of 25% of the total floor area. As calculated, the total area of openings is 26.8% of the total floor area meaning it is not complied as it doesn't fit the criteria.

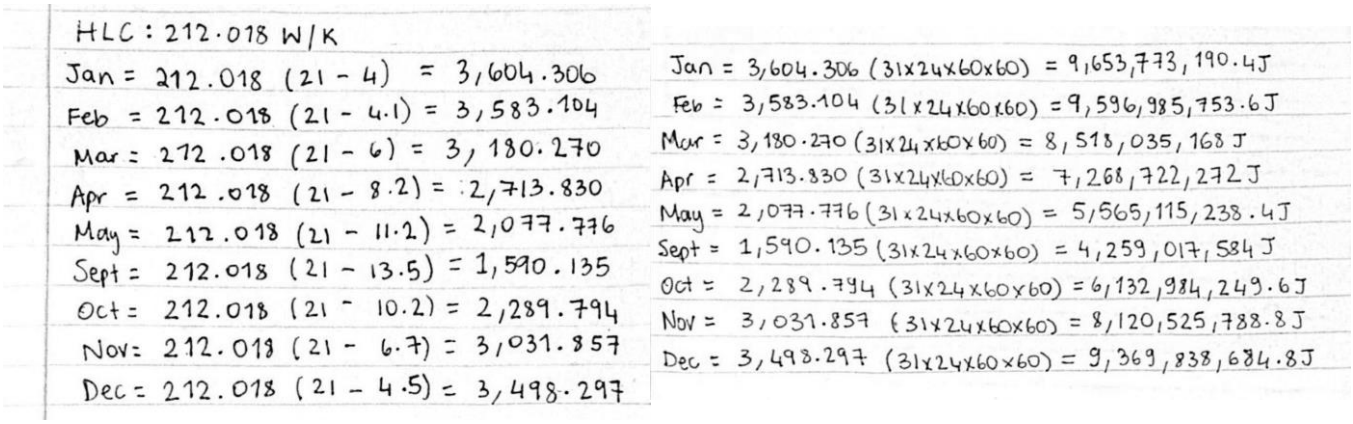
**Building Fabric Heat Loss and Annual Energy Demand 1,962**

| Building Element | Area ($m^2$) | U-Value ($m^2$ .K/W) | HLC (A x U) (W/K) |
|---|---|---|---|
| Floor | 82.808 | 0.600 | 49.685 |
| Wall | 80.673 | 0.576 | 46.468 |
| Roof | Area of ceiling= 82.808 | 0.666 | 55.150 |
| Window | 17.443 | 2.800 | 48.840 |
| Door | 4.750 | 2.500 | 11.875 |
| | | Total: | 212.018 |

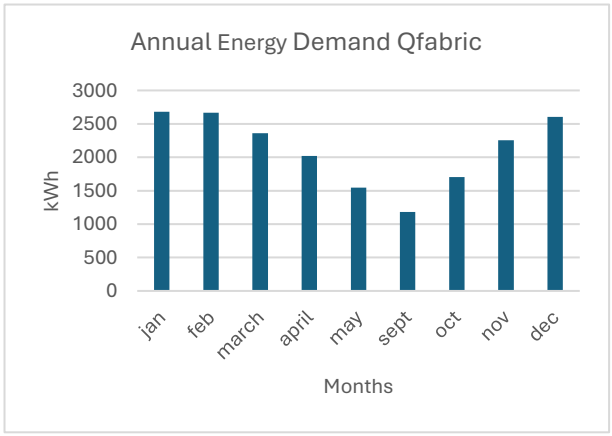
- The roof loses the most heat with an HLC of 55.150 W/K due to its large area and high U-value (0.666), indicating insufficient thermal resistance.
- The floor follows at 49.685 W/K, with a lower U-value (0.600), but its large area and limited insulation contribute to heat loss.
- Walls have an HLC of 46.468 W/K; despite a relatively low U-value (0.576), their large area and lack of insulation increase heat loss.
- Windows, with the highest U-value (2.800), contribute an HLC of 48.840 W/K, showing poor insulation due to likely single glazing or low-performance glazing.
- The door contributes the least heat loss (11.875 W/K), with a high U-value (2.500) but minimal impact due to its small size.

With a combined HLC of 212.018 W/K, the building shows significant thermal inefficiency, with the roof and windows as key areas for improvement. Adding insulation and upgrading glazing could reduce heat loss.

**Annual Energy Demand and $CO_2$ emissions just from fabric heat losses**




| Joules (J) | kWh |
|---|---|
| 9653773190.4 | 2681.60 |
| 9596985753.6 | 2665.83 |
| 8518035168.0 | 2360.57 |
| 7268722272.0 | 2018.03 |
| 5565115238.4 | 1545.87 |
| 4259017584.0 | 1183.61 |
| 6132984249.6 | 1703.05 |
| 8120525788.8 | 2255.81 |
| 9369838684.8 | 2605.54 |
| **Total = 68,484,997,929.60 J** | **Total = 19,019.91 kwh** |



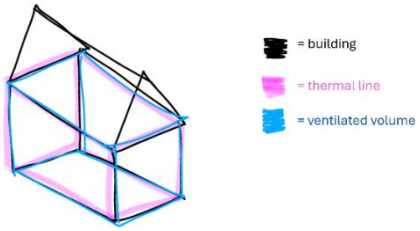

The total annual energy demand is 19,019.91 kwh.

Fabric heat loss fluctuates with outside temperature, varying monthly. Energy loss is calculated excluding summer months (June-August) when heating isn't needed. This reflects the energy lost per second each month, requiring replenishment to maintain a 21°C indoor temperature for client comfort.

**Potential additional issues:**

- An unventilated roof can cause warm air to cool, leading to condensation, fabric deterioration, wood rot, and mold growth, which pose health risks.
- Insulation between joists causes thermal bridging, allowing heat to escape, increasing energy costs to maintain a comfortable temperature.
- Double-glazed windows with low thermal performance allow cold air in, making it harder to maintain indoor temperature and raising energy costs.
- In rainy areas like Nottingham, unventilated wall cavities can trap moisture, damaging the building and promoting mold growth.
- Timber frames for windows may rot or warp if exposed to rain, requiring protective coating for durability.

## Ventilation




Shown in blue is the ventilated volume of the building.

There is a system for purge ventilation in every room of the house. Every bathroom, bedroom and living space has a window, the hallway has a main door to the outside. In the kitchen there is a cooker hood at an extract rate of low setting 288m3/hr and high setting 432m3/hr. In the bathroom there is a bathroom fan at an extract rate of 82.8m3/hr plus 10-minute overrun timer every time it is switched on.

These systems manage moisture, pollutants, and odours effectively, but performance depends on size, placement, and cross-ventilation. While mechanical systems meet ventilation needs, they increase energy use. Improving natural ventilation by meeting the 5% floor area rule, optimizing window placement, and adding passive features like trickle vents or transom windows can reduce energy costs and boost efficiency.

## Air change rate needed and Adventitious infiltration

Adventitious infiltration rate = 1.1 $ach^{-1}$

ventilated volume of the house = 2.45 x 6.993 x 14
= 239.86 $m^3$

what air change rate do I need?

Approved Document F ventilation rate for 3 bedrooms = 31 l/s
0.3 l/s per $m^2$ ventilation rate
0.3 x (total area of 3 bedrooms)
0.3 x 29.52 = 8.856

use 31 l/s as Part F suggests use the highest ventilation rate for 3 bedrooms.

$q\ (m^3/s)$ = whole dwelling ventilation rate / 1000

31 / 1000 = 0.031 $m^3/s$

air change rate = $n = \frac{q \times 3600}{V}$

$= \frac{0.031 \times 3600}{239.86} = 0.465\ ach^{-1}$

air change needed = 0.47 $ach^{-1}$
adventitious infiltration rate = 1.1 $ach^{-1}$ over-ventilating the space.

The required air change rate is 0.47, which is sufficient to provide adequate ventilation for the occupants. However, the adventitious infiltration, which refers to unwanted air entering the building, is measured at 1.1. This indicates that the house is significantly overventilated, as 1.1 exceeds the required 0.47. This overventilation is likely caused by cracks and gaps in the construction. As a result, additional energy is lost, which must be replaced with thermal energy to maintain the interior at the client's preferred comfort temperature. This increased energy consumption also contributes to higher CO2 emissions

## Air permeability

5 $m^3 / (h \times m^2)$ @ 50 Pa
House should loose no more than 5 $m^3$ of every hour for every 1 $m^2$ of external surface area.
Surface area = air tightness line
Ceiling = 82.808 $m^2$
floor = 82.808 $m^2$
walls = 80.673
Total (SA) = 82.808 + 82.808 + 80.673 = 246.289
in an hour the house should lose no more than
246.289 x 5 = 1231.445 $m^3$ @ 50 Pa

$n = \frac{1231.445}{239.86} = 5.134\ ach^{-1}$

$n_{50} - 5.134\ ach^{-1} = \frac{5.134}{20.6} = 0.249\ ach^{-1}$

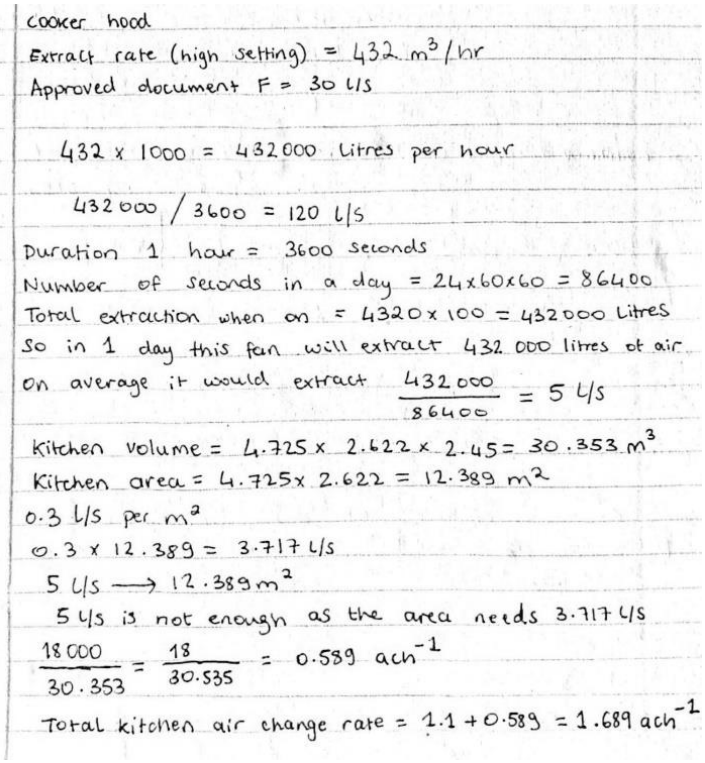
adventitious rate should be no more than 0.249 $ach^{-1}$
adventitious ventilation rate = 1.1 $ach^{-1}$
↳ significantly over-ventilating the space.

The calculations show the building's airflow, with a desired adventitious rate of 0.249 and an actual infiltration rate of 1.1. This indicates:

- Over-ventilation, with excess air entering the building.
- Insufficient airtightness, allowing cold air in and warm air out, increasing energy use to maintain the desired temperature.

These gaps may also let rainwater in, causing dampness and mold, which could harm the client's health.

## Cooker Hood



cooker hood
Extract rate (high setting) = 432 $m^3$/hr
Approved document F = 30 l/s

432 x 1000 = 432000 Litres per hour

432000 / 3600 = 120 l/s

Duration 1 hour = 3600 seconds
Number of seconds in a day = 24x60x60 = 86400
Total extraction when on = 4320 x 100 = 432000 Litres
So in 1 day this fan will extract 432 000 litres of air
On average it would extract $\frac{432000}{86400} = 5$ l/s

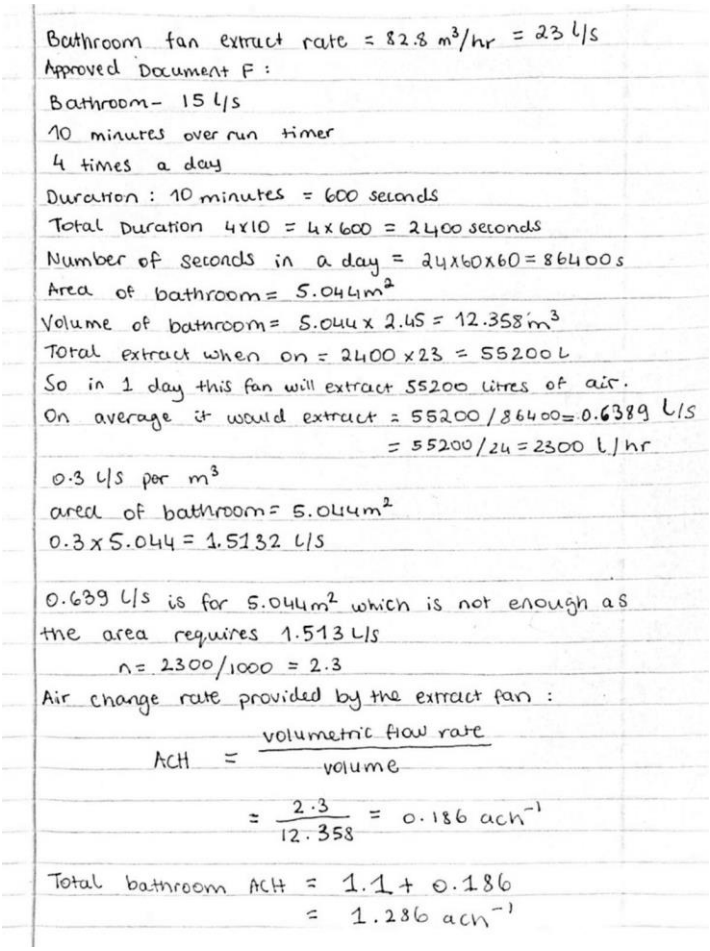
Kitchen volume = 4.725 x 2.622 x 2.45 = 30.353 $m^3$
Kitchen area = 4.725 x 2.622 = 12.389 $m^2$
0.3 l/s per $m^2$
0.3 x 12.389 = 3.717 l/s
5 l/s ⟶ 12.389 $m^2$
5 l/s is not enough as the area needs 3.717 l/s
$\frac{18000}{30.353} = \frac{18}{30.535} = 0.589$ $ach^{-1}$
Total kitchen air change rate = 1.1 + 0.589 = 1.689 $ach^{-1}$

Approved Document F shows that the extract rate for a kitchen should be a minimum of 30l/s. The cooker hood extract rate at the maximum setting is 432 $m^3$/hr= 120l/s.

The cooker hood meets the Approved Document F requirements as 120l/s > 30l/s.

The cooker hood extracts 5 l/s, exceeding the required 3.717 l/s for the kitchen. This over-extraction leaves the room under-ventilated and lacking sufficient air supply for client comfort. Therefore, the cooker hood is inadequate as a sole mechanical ventilation system at its maximum setting.

## Bathroom Extract Fan



Bathroom fan extract rate = 82.8 $m^3$/hr = 23 l/s
Approved Document F:
Bathroom- 15 l/s
10 minutes over run timer
4 times a day
Duration: 10 minutes = 600 seconds
Total Duration 4x10 = 4x600 = 2400 seconds
Number of seconds in a day = 24x60x60 = 86400 s
Area of bathroom = 5.044 $m^2$
Volume of bathroom = 5.044 x 2.45 = 12.358 $m^3$
Total extract when on = 2400 x 23 = 55200 L
So in 1 day this fan will extract 55200 litres of air.
On average it would extract = 55200 / 86400 = 0.6389 l/s
= 55200 / 24 = 2300 l/hr
0.3 l/s per $m^3$
area of bathroom = 5.044 $m^2$
0.3 x 5.044 = 1.5132 l/s

0.639 l/s is for 5.044 $m^2$ which is not enough as the area requires 1.513 l/s
n = 2300 / 1000 = 2.3
Air change rate provided by the extract fan:

$$ACH = \frac{\text{volumetric flow rate}}{\text{volume}}$$

$$= \frac{2.3}{12.358} = 0.186 \text{ ach}^{-1}$$

Total bathroom ACH = 1.1 + 0.186
= 1.286 $ach^{-1}$

Approved Document F specifies that the required extract rate for a bathroom is 15 L/s. The fan's extract rate is 23 L/s, which meets this requirement as 23 L/s > 15 L/s.

The fan extracts air at 0.639 L/s, far below the required 1.513 L/s for the bathroom, leading to inadequate ventilation. This makes the fan insufficient as a mechanical ventilation system.

**HLC**ventilation **and Q**ventilation 1,023

Natural ventilation

HLC = 1/3 x n x V  n = 1.1 ach$^{-1}$

= 1/3 x 1.1 x 239.86

= 87.94867

Qv:

Jan = 87.95 (21−4) = 1495.15

Feb = 87.95 (21−4.1) = 1486.355

Mar = 87.95 (21 − 6) = 1319.25

Apr = 87.95 (21−8.2) = 1125.76

May = 87.95 (21 − 11.2) = 861.91

Sep = 87.95 (21 − 13.5) = 659.625

Oct = 87.95 (21 − 10.2) = 949.86

Nov = 87.95 (21 − 6.7) = 1257.685

Dec = 87.95 (21 − 4.5) = 1451.175

Cooker hood ventilation

HLC = 1/3 x n x V  n (total) = 1.689 ach$^{-1}$

= 1/3 x 1.689 x 30.853

= 17.0887

Qv:

Jan = 17.089 (21−4) = 290.513

Feb = 17.089 (21−4.1) = 288.804

Mar = 17.089 (21−6) = 256.335

Apr = 17.089 (21−8.2) = 218.739

May = 17.089 (21−11.2) = 167.472

Sep = 17.089 (21 − 13.5) = 128.168

Oct = 17.089 (21 − 10.2) = 184.561

Nov = 17.089 (21 − 6.7) = 244.373

Dec = 17.089 (21 − 4.5) = 281.969

Bathroom ventilation

HLC = 1/3 x n x V  n (total) = 1.286 ach$^{-1}$

= 1/3 x 1.286 x 12.358

= 5.2975

Qv:

Jan = 5.298 (21 − 4) = 90.066

Feb = 5.298 (21 − 4.1) = 89.536

Mar = 5.298 (21 − 6) = 79.47

Apr = 5.298 (21 − 8.2) = 67.814

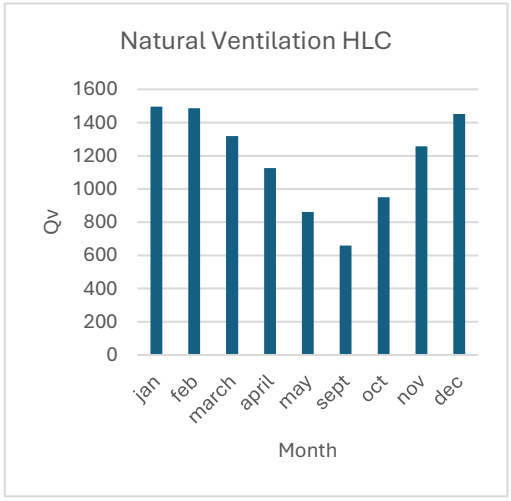
May = 5.298 (21 − 11.2) = 51.92

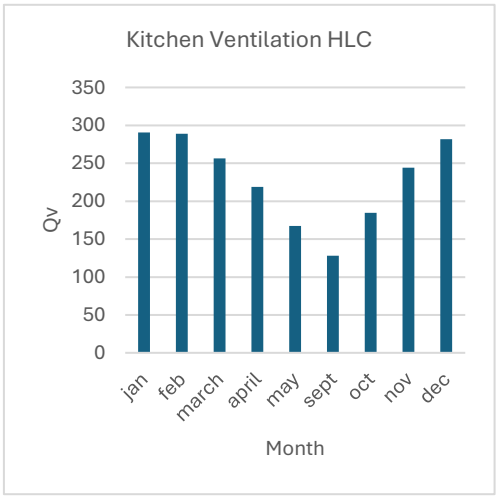
Sep = 5.298 (21 − 13.5) = 39.735

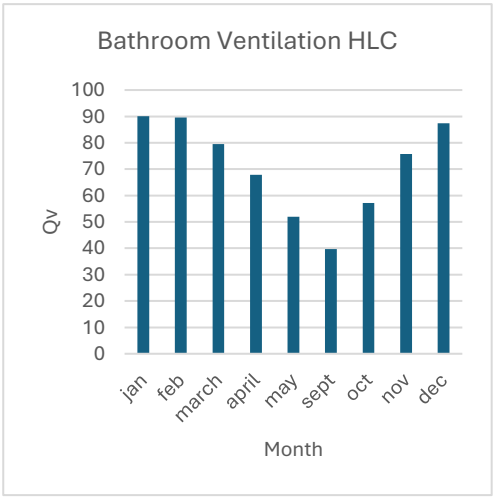
Oct = 5.298 (21 − 10.2) = 57.218

Nov = 5.298 (21 − 6.7) = 75.761

Dec = 5.298 (21 − 4.5) = 87.417




Thermal energy loss from natural and mechanical ventilation is calculated based on variations in outdoor temperature, as these fluctuations affect the temperature difference, causing heat loss to change each month.

- Natural Ventilation Losses: Represent the heat lost through gaps and openings in the building's construction.
- Mechanical Ventilation Losses: Account for the heat lost due to the operation of the bathroom extractor fan and cooker hood, which remove warm air from these areas.

**Energy per month and Annual energy demand per month due to ventilation losses**

Jan = (1495.15 + 290.513 + 90.066) x (31 x 24 x 60 x 60) = 5,023,952,553.6 J = 1,395.54 kWh

Feb = (1486.355 + 288.804 + 89.536) x (28 x 24 x 60 x 60) = 4,511,070,144 J = 1,253.08 kWh

Mar = (1319.25 + 256.335 + 79.47) x (31 x 24 x 60 x 60) = 4,432,899,312 J = 1,231.36 kWh

Apr = (1125.76 + 218.739 + 67.814) x (30 x 24 x 60 x 60) = 3,660,715,296 J = 1,016.87 kWh

May = (861.91 + 167.472 + 51.92) x (31 x 24 x 60 x 60) = 2,896,159,276.8 J = 804.49 kWh

Sep = (659.625 + 128.168 + 39.735) x (30 x 24 x 60 x 60) = 2,144,952,576 J = 595.32 kWh

Oct = (949.86 + 184.561 + 57.218) x (31 x 24 x 60 x 60) = 3,191,635,897.6 J = 886.58 kWh

Nov = (1257.685 + 244.373 + 75.761) x (30 x 24 x 60 x 60) = 4,089,706,848 J = 1136.03 kWh

Dec = (1451.175 + 281.969 + 87.417) x (31 x 24 x 60 x 60) = 4,876,190,582.4 J = 1354.497 kWh

Total = 348,273,324,86.4 J

Annual Energy Demand = 348,273,324,86.4 / 360 0000

= 9,674.25 kWh

Based on $\mathbf{Q}_{\text{natural ventilation}}$ and $\mathbf{Q}_{\text{mechanical ventilation}}$, monthly energy loss is calculated, excluding the summer months (June, July, and August), as heating is not needed during this period, meaning energy loss does not require replacement. This loss represents the energy lost through ventilation every second over the course of a month, which needs to be replenished to maintain an optimal interior temperature of 21°C. When compared to fabric heat loss, the ventilation loss is significantly smaller.

## Heat Gains

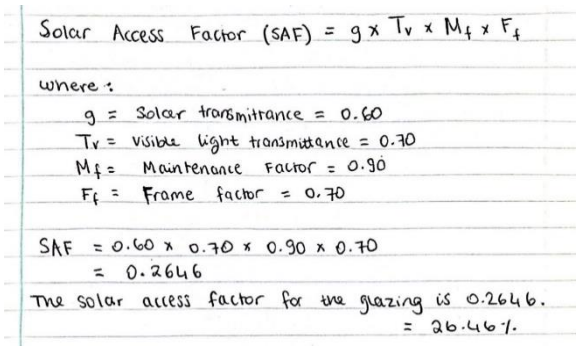
### Fixed Appliances

| Room | Type | Rating (W) | Time in use (s) | Energy (Joules) |
|---|---|---|---|---|
| Living room | LCD TV<br>Games Console<br>Broadband Router<br>Tablet charger<br>Smartphone charger | 71<br>100<br>14<br>10<br>5 | 3 hours= 10800<br>1 hour= 3600<br>24 hours= 86400<br>1 hour= 3600<br>3 hours= 10800 | 766,800<br>360,000<br>1,209,600<br>36,000<br>54,000 |
| Kitchen | Oven (electric)<br>Hob (gas)<br>Cooker hood<br>Fridge freezer<br>Dishwasher<br>Microwave<br>Kettle<br>Toaster<br>Air Fryer | 3100<br>2600<br>110<br>260kWh PA = 29.7 W<br>1.45kW= 1450W<br>800<br>3050<br>850<br>2470 | 30 minutes= 1800<br>1 hour= 3600<br>1 hour= 3600<br>24 hours= 86400<br>1 hour= 3600<br>30 minutes= 1800<br>10 minutes= 600<br>10 minutes = 600<br>30 minutes = 1800 | 5,580,000<br>9,360,000<br>396,000<br>2,566,080<br>5,220,000<br>1,440,000<br>1,830,000<br>510,000<br>4,446,000 |
| Bathroom | Extractor Fan<br>Hairdryer | 42<br>1600 | 40 minutes= 2400<br>10 minutes= 600 | 100,800<br>960,000 |
| Bedroom 1 | TV<br>Electric blanket<br>Smartphone charger | 60<br>110<br>5 | 2 hours= 7200<br>3 hours= 10800<br>3 hours= 10800 | 432,000<br>1,188,000<br>54,000 |
| Bedroom 2 | TV<br>Games console<br>Smartphone charger | 40<br>100<br>5 | 2 hours= 7200<br>2 hours= 7200<br>3 hours= 10800 | 288,000<br>720,000<br>54,000 |
| Bedroom 3 | Electric blanket<br>Smartphone charger | 110<br>5 | 3 hours = 10800<br>3 hours = 10800 | 1,188,000<br>54,000 |
| | | | Total: | 38,813,280 |

Mean heat gain from appliances: **Q**A=61,673,280/24 x 60 x 60 = 713.811 W
This is the total thermal energy gained per second per day just from fixed appliances.

The table provides an estimate of the appliances the clients are likely to use, along with their expected usage durations. The washing machine and dishwasher have the highest thermal gain contribution, whereas the smartphone and tablet chargers contribute the least.

### Solar




Calculation for Solar Access Factor (SAF) showing how much sunlight the glazing receives when there is partial shade/obstruction. Only one façade (South Elevation) of the building is obstructed by an identical building where 1 window

will receive little sunlight, that value will be assumed as Z=0.5 as the obstruction is identical in dimension, meaning with the sun path it is partially obstructed.

**Solar Gains**

Equation = 0.9 x Transmittance Factor (gi) x Solar Radiation (S) x Solar Access Factor (Z) x Area of openings x Frame Factor (FF) = Solar Heat Gains Qs(W)

Transmittance Factor = 0.60 as per brief

Solar radiation = calculated using spreadsheet 01, inputting the orientation in degrees

Solar access factor = Z = 1 if no obstruction and Z = 0.5 partial obstruction

Area of opening = Shown in table below

Frame Factor = Ratio glass/window frame = 0.70 as per brief

| | Living Room | Kitchen | W.C. | Bathroom | Bedroom 1 | Bedroom 2 | Bedroom 3 | |
|---|---|---|---|---|---|---|---|---|
| | Window (East) | Window (WEST) | Window (EAST) | Window (WEST) | Window (SOUTH) | Window (EAST) | Window (WEST) | |
| **Solar Access (Z,winter)** | 1 | 1 | 1 | 1 | 0.5 | 1 | 1 | |
| **g** | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | |
| **FF** | 0.7 | 0.7 | 0.7 | 0.7 | 0.7 | 0.7 | 0.7 | |
| **Area (m2)** | 3.55 | 1.61 | 0.19 | 0.77 | 1.83 | 3.55 | 0.77 | |
| | | | | **Qsolar(W)** | | | | **Total** |
| **Jan** | 26.663553 | 12.0940454 | 1.4270634 | 5.78410867 | 32.7331368 | 26.66699452 | 5.78410867 | 111.153 |
| **Feb** | 51.689988 | 23.44169667 | 2.7665064 | 11.21124623 | 53.3884932 | 51.68821315 | 11.21124623 | 205.397 |
| **Mar** | 82.620783 | 37.46737824 | 4.4219574 | 17.9191809 | 65.196495 | 82.61440543 | 17.9191809 | 308.159 |
| **Apr** | 122.663079 | 55.63016635 | 6.5650662 | 26.60573173 | 72.7087914 | 122.6627891 | 26.60573173 | 433.441 |
| **May** | 149.246118 | 67.68607553 | 7.9878204 | 32.37160134 | 75.088377 | 149.2456945 | 32.37160134 | 513.997 |
| **Sep** | 98.76384 | 44.79378174 | 5.285952 | 21.42311301 | 69.1670826 | 98.76889762 | 21.42311301 | 359.626 |
| **Oct** | 62.948529 | 28.54758021 | 3.3690762 | 13.65319054 | 58.9985046 | 62.9465278 | 13.65319054 | 244.117 |
| **Nov** | 33.158349 | 15.03603886 | 1.7746722 | 7.191149018 | 38.78517006 | 33.15399872 | 7.191149018 | 136.291 |
| **Dec** | 21.993741 | 9.976394587 | 1.1771298 | 4.771319151 | 28.2852486 | 21.99764024 | 4.771319151 | 92.9728 |

Using the given spreadsheet 01 and 02, I calculated the solar gains per window in each room. Cooling periods where heating isn't on in June, July, and August are excluded from the table as the values are 0. Summer months are excluded as heating isn't required during those months; it is not included in the total thermal gains.
**Down below shows Q loss from glazing.**

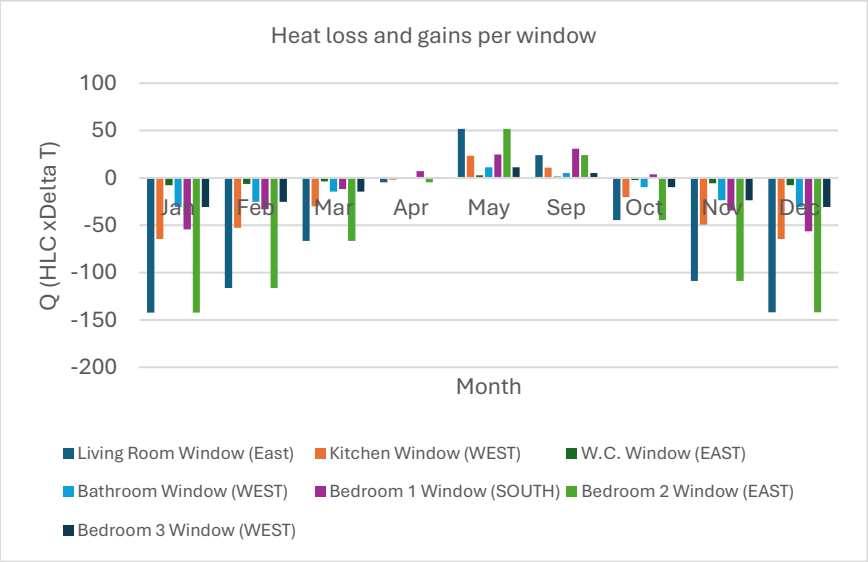
| | | | | Living Room | Kitchen | W.C. | Bathroom | Bedroom 1 | Bedroom 2 | Bedroom 3 |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Window (East) | Window (WEST) | Window (EAST) | Window (WEST) | Window (SOUTH) | Window (EAST) | Window (WEST) |
| | | | **Area (m2)** | 3.55 | 1.61 | 0.19 | 0.77 | 1.83 | 3.55 | 0.77 |
| | **Delta T** | **U-value (W/m2K)** | | **Q loss from glazing, W (HLC x Delta T)** | | | | | | |
| **Jan** | 17 | 2.8 | | 168.98 | 76.636 | 9.044 | 36.652 | 87.108 | 168.98 | 36.652 |
| **Feb** | 16.9 | 2.8 | | 167.986 | 76.1852 | 8.9908 | 36.4364 | 86.5956 | 167.986 | 36.4364 |
| **Mar** | 15 | 2.8 | | 149.1 | 67.62 | 7.98 | 32.34 | 76.86 | 149.1 | 32.34 |
| **Apr** | 12.8 | 2.8 | | 127.232 | 57.7024 | 6.8096 | 27.5968 | 65.5872 | 127.232 | 27.5968 |
| **May** | 9.8 | 2.8 | | 97.412 | 44.1784 | 5.2136 | 21.1288 | 50.2152 | 97.412 | 21.1288 |
| **Sep** | 7.5 | 2.8 | | 74.55 | 33.81 | 3.99 | 16.17 | 38.43 | 74.55 | 16.17 |
| **Oct** | 10.8 | 2.8 | | 107.352 | 48.6864 | 5.7456 | 23.2848 | 55.3392 | 107.352 | 23.2848 |
| **Nov** | 14.3 | 2.8 | | 142.142 | 64.4644 | 7.6076 | 30.8308 | 73.2732 | 142.142 | 30.8308 |
| **Dec** | 16.5 | 2.8 | | 164.01 | 74.382 | 8.778 | 35.574 | 84.546 | 164.01 | 35.574 |

This represents the heat loss through the same glazing.

**Q gains and loses**

| | Living Room | Kitchen | W.C. | Bathroom | Bedroom 1 | Bedroom 2 | Bedroom 3 |
|---|---|---|---|---|---|---|---|
| | Window (East) | Window (WEST) | Window (EAST) | Window (WEST) | Window (SOUTH) | Window (EAST) | Window (WEST) |
| **Jan** | -142.316447 | -64.5419546 | -7.6169366 | -30.86789133 | -54.3748632 | -142.3130055 | -30.86789133 |
| **Feb** | -116.296012 | -52.74350333 | -6.2242936 | -25.22515377 | -33.2071068 | -116.2977869 | -25.22515377 |
| **Mar** | -66.479217 | -30.15262176 | -3.5580426 | -14.4208191 | -11.663505 | -66.48559457 | -14.4208191 |
| **Apr** | -4.568921 | -2.072233653 | -0.2445338 | -0.991068269 | 7.1215914 | -4.56921085 | -0.991068269 |
| **May** | 51.834118 | 23.50767553 | 2.7742204 | 11.24280134 | 24.873177 | 51.8336945 | 11.24280134 |
| **Sep** | 24.21384 | 10.98378174 | 1.295952 | 5.253113005 | 30.7370826 | 24.21889762 | 5.253113005 |
| **Oct** | -44.403471 | -20.13881979 | -2.3765238 | -9.631609464 | 3.6593046 | -44.4054722 | -9.631609464 |
| **Nov** | -108.983651 | -49.42836114 | -5.8329278 | -23.63965098 | -34.48802994 | -108.9880013 | -23.63965098 |
| **Dec** | -142.016259 | -64.40560541 | -7.6008702 | -30.80268085 | -56.2607514 | -142.0123598 | -30.80268085 |
| | | | | | | | |
| | **positive is gain** | | | | | | |
| | **negative is loss** | | | | | | |

Graph showing the heat losses and gains per month, each bar representing a specific window.



## Occupancy Schedule

| Activity | Room where it takes place | Parent 1 | Time spent in each room (min) | Parent 2 | Time spent in each room (min) |
|---|---|---|---|---|---|
| Waking up | Bedroom | 06:30 | 515 (8.5h) | 06:30 | 515 (8.5h) |
| Shower | Bathroom | 6:30-6:45 | 15 | 6:30-6:45 | 15 |
| Breakfast | Kitchen | 6:45-7:00 | 15 | 6:45-7:00 | 15 |
| Leave for work/school | / | 7:00-17:00 | / | 7:00-17:00 | / |
| Do HW with kids | Living room | 17:00-18:00 | 120 | 17:00-18:00 | 120 |
| Chores | Kitchen | 18:00-19:00 | 60 | 18:00-19:00 | 60 |
| Dinner | Kitchen | 19:00 - 20:00 | 60 | 19:00 - 20:00 | 60 |
| Reads/Watches TV | Living Room | 20:00-22:00 | 60 | 20:00-22:00 | 60 |
| Sleep | Bedroom | 22:00-6:30 | / | 22:00-6:30 | / |
| **Activity** | **Room where it takes place** | **Child 1** | **Time spent in each room (min)** | **Child 2** | **Time spent in each room (min)** |
| Waking up | Bedroom | 6:55-7:00 | 5 | 6:55-7:00 | 5 |
| Shower | Bathroom | 7:00-7:15 | 15 | 7:00-7:15 | 15 |
| Breakfast | Kitchen | 7:15-7:45 | 30 | 7:15-7:45 | 30 |
| Leave for school | / | 7:45-16:00 | / | 7:45-16:00 | / |
| Do HW | Living room | 16:00-18:00 | 120 | 16:00-18:00 | 120 |
| Free Time | Bedroom | 18:00-19:00 | 60 | 18:00-19:00 | 60 |
| Dinner | Kitchen | 19:00 - 20:00 | 60 | 19:00 - 20:00 | 60 |
| Sleep | Bedroom | 20:00 | / | 20:00 | / |

Occupancy schedule showing how the day of an average family might occur.

## Metabolic Heat Gains

Metabolic Heat Gains = Activity level x Time Occupied

Assuming that they were sitting at rest the whole time, I assumed their activity level as 115W for a male and applied the correction factor of 0.85 for the female (making 97.75W) and 0.75 for the children (making 86.25). These calculations will help understand the heat gained from their daily activities in the house.

| **Bedroom 1** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Parent 1 | 115 | 8.5 | 30600 | 3519000 |
| Parent 2 | 97.75 | 8.5 | 30600 | 3519000 |
| | | | **Total (J)** | 7038000 |
| | | | **Total (W)** | **81.46** |

| **Bedroom 2** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Child 1 | 86.25 | 11 | 39600 | 3415500 |
| | | | **Total (J)** | 3415500 |
| | | | **Total (W)** | **39.53** |

| **Bedroom 3** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Child 2 | 86.25 | 11 | 39600 | 3415500 |
| | | | **Total (J)** | 3415500 |
| | | | **Total (W)** | **39.53** |

| **Bathroom** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Parent 1 | 115 | 0.25 | 900 | 103500 |
| Parent 2 | 97.75 | 0.25 | 900 | 87975 |
| Child 1 | 86.25 | 0.25 | 900 | 77625 |
| Child 2 | 86.25 | 0.25 | 900 | 77625 |
| | | | **Total (J)** | 346725 |
| | | | **Total (W)** | **4.01** |

| **Living Room** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Parent 1 | 115 | 3 | 10800 | 1242000 |
| Parent 2 | 97.75 | 3 | 10800 | 1055700 |
| Child 1 | 86.25 | 2 | 7200 | 621000 |
| Child 2 | 86.25 | 2 | 7200 | 621000 |
| | | | **Total (J)** | 3539700 |
| | | | **Total (W)** | **40.97** |

| **Kitchen** | | | | |
|---|---|---|---|---|
| **Occupant** | **Activity Level (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Parent 1 | 115 | 2.25 | 8100 | 931500 |
| Parent 2 | 97.75 | 2.25 | 8100 | 791775 |
| Child 1 | 86.25 | 1.5 | 5400 | 465750 |
| Child 2 | 86.25 | 1.5 | 5400 | 465750 |
| | | | **Total (J)** | 2654775 |
| | | | **Total (W)** | **30.73** |

**Total Metabolic Gains Average over a one second snap shot of the day (W)**

**Total : 236.23**

## Lighting

Using the occupancy schedule, the sunrise and sunset times. I can calculate how many minutes the light would be on during one whole day for each month for each room.

| | Living Room | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |
| Feb | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |
| Mar | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |
| Apr | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |
| May | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 3 | 10800 | 2808000 | 32.50 |
| Sep | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 3 | 10800 | 2808000 | 32.50 |
| Oct | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 3 | 10800 | 2808000 | 32.50 |
| Nov | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |
| Dec | 5 | 100 W for 1 light, 40W for 4 lights | 260 | 4 | 14400 | 3744000 | 43.33 |

| | Bedroom 1 | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |
| Feb | 3 | 1x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |
| Mar | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |
| Apr | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |
| May | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 2 | 7200 | 1008000 | 11.67 |
| Sep | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 2 | 7200 | 1008000 | 11.67 |
| Oct | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 2 | 7200 | 1008000 | 11.67 |
| Nov | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |
| Dec | 3 | 1 x 60W incandescent, 2 x 40W wall lights | 140 | 3 | 10800 | 1512000 | 17.50 |

| | Bedroom 2 | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Feb | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Mar | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Apr | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| May | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Sep | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Oct | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Nov | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Dec | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |

| | Bedroom 3 | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Feb | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Mar | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Apr | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| May | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Sep | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Oct | 1 | 60 | 60 | 2 | 7200 | 432000 | 5.00 |
| Nov | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |
| Dec | 1 | 60 | 60 | 3 | 10800 | 648000 | 7.50 |

| | Kitchen | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 6 | 50 | 300 | 4 | 14400 | 4320000 | 50.00 |
| Feb | 6 | 50 | 300 | 4 | 14400 | 4320000 | 50.00 |
| Mar | 6 | 50 | 300 | 4 | 14400 | 4320000 | 50.00 |
| Apr | 6 | 50 | 300 | 4 | 14400 | 4320000 | 50.00 |
| May | 6 | 50 | 300 | 4 | 14400 | 4320000 | 50.00 |
| Sep | 6 | 50 | 300 | 3 | 10800 | 3240000 | 37.50 |
| Oct | 6 | 50 | 300 | 3 | 10800 | 3240000 | 37.50 |
| Nov | 6 | 50 | 300 | 3 | 10800 | 3240000 | 37.50 |
| Dec | 6 | 50 | 300 | 3 | 10800 | 3240000 | 37.50 |

| | Bathroom | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| Feb | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| Mar | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| Apr | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| May | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| Sep | 4 | 50 | 200 | 1 | 3600 | 720000 | 8.33 |
| Oct | 4 | 50 | 200 | 1 | 3600 | 720000 | 8.33 |
| Nov | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |
| Dec | 4 | 50 | 200 | 2 | 7200 | 1440000 | 16.67 |

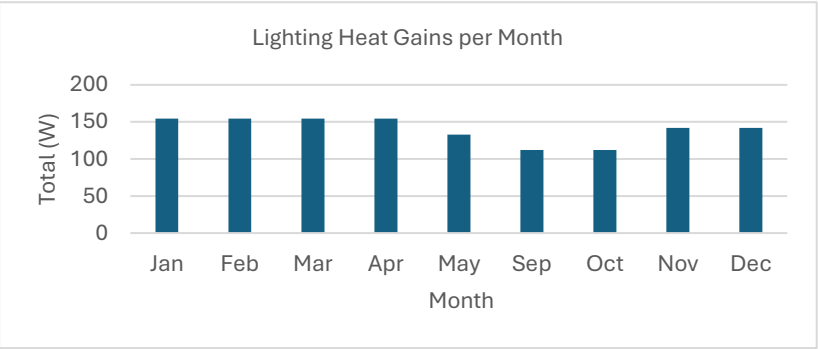
| | W.C. | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Feb | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Mar | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Apr | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| May | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Sep | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Oct | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Nov | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |
| Dec | 1 | 50 | 50 | 1 | 3600 | 180000 | 2.08 |

| | Hallway | | | | | | |
|---|---|---|---|---|---|---|---|
| | Number of Lights | Power consumption per light (W) | Total power consumption (W) | Daily Usage (h) | Daily usage (s) | Total daily energy consumption (J) | Average W (1 second) |
| Jan | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Feb | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Mar | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Apr | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| May | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Sep | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Oct | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Nov | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |
| Dec | 3 | 40 | 120 | 2 | 7200 | 864000 | 10.00 |

From the tables above we can deduce the following:

| | Living Room | Bedroom 1 | Bedroom 2 | Bedroom 3 | Kitchen | Bathroom | W.C. | Hallway | TOTAL |
|---|---|---|---|---|---|---|---|---|---|
| **Jan** | 43.33 | 17.5 | 7.5 | 7.5 | 50 | 16.67 | 2.08 | 10 | **154.6** |
| **Feb** | 43.33 | 17.5 | 7.5 | 7.5 | 50 | 16.67 | 2.08 | 10 | **154.6** |
| **Mar** | 43.33 | 17.5 | 7.5 | 7.5 | 50 | 16.67 | 2.08 | 10 | **154.6** |
| **Apr** | 43.33 | 17.5 | 7.5 | 7.5 | 50 | 16.67 | 2.08 | 10 | **154.6** |
| **May** | 32.5 | 11.67 | 5 | 5 | 50 | 16.67 | 2.08 | 10 | **132.9** |
| **Sep** | 32.5 | 11.67 | 5 | 5 | 37.5 | 8.33 | 2.08 | 10 | **112.1** |
| **Oct** | 32.5 | 11.67 | 5 | 5 | 37.5 | 8.33 | 2.08 | 10 | **112.1** |
| **Nov** | 43.33 | 17.5 | 7.5 | 7.5 | 37.5 | 16.67 | 2.08 | 10 | **142.1** |
| **Dec** | 43.33 | 17.5 | 7.5 | 7.5 | 37.5 | 16.67 | 2.08 | 10 | **142.1** |

This then can give us a total average power consumption rating for the lights in the whole house per year of **1260W**. Here you can clearly see that as the months get colder and darker the heat gains increase due to the high need for light and as the month get lighter heat gains decrease.




Graph showing monthly total lighting gains, January through April being the highest and September and October being the lowest.

## Appliance Heat Gains:

| **Bedroom 1** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| TV | 60 | 2 | 7200 | 432000 |
| Electric blanket | 110 | 3 | 10800 | 1188000 |
| Smartphone charger | 5 | 3 | 10800 | 54000 |
| | | | **Total (J)** | 1674000 |
| | | | **Total (W)** | **19.375** |

| **Bedroom 2** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| TV | 40 | 2 | 7200 | 288000 |
| Games console | 100 | 2 | 7200 | 720000 |
| Smartphone charger | 5 | 3 | 10800 | 54000 |
| | | | **Total (J)** | 1062000 |
| | | | **Total (W)** | **12.2917** |

| **Bedroom 3** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Electric blanket | 110 | 3 | 10800 | 1188000 |
| Smartphone charger | 5 | 3 | 10800 | 54000 |
| | | | **Total (J)** | 1242000 |
| | | | **Total (W)** | **14.375** |

| **Kitchen** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Oven (electric) | 3100 | 0.5 | 1800 | 5580000 |
| Hob (gas) | 2600 | 1 | 3600 | 9360000 |
| Cooker hood | 110 | 1 | 3600 | 396000 |
| Fridge freezer | 29.7 | 24 | 86400 | 2566080 |
| Dishwasher | 1450 | 1 | 3600 | 5220000 |
| Mircowave | 800 | 0.5 | 1800 | 1440000 |
| Kettle | 3050 | 0.166 | 600 | 1830000 |
| Toaster | 850 | 0.166 | 600 | 510000 |
| Air Fryer | 2470 | 0.5 | 1800 | 4446000 |
| | | | **Total (J)** | 31348080 |
| | | | **Total (W)** | **362.825** |

| **Living room** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| LCD TV | 71 | 3 | 10800 | 766800 |
| Games console | 100 | 1 | 3600 | 360000 |
| Broadband router | 14 | 24 | 86400 | 1209600 |
| Tablet Charger | 10 | 1 | 3600 | 36000 |
| Smartphone charger | 5 | 3 | 10800 | 54000 |
| | | | **Total (J)** | 2426400 |
| | | | **Total (W)** | **28.0833** |

| **Bathroom** | | | | |
|---|---|---|---|---|
| **Appliance** | **Power (W)** | **Time Occupied (h)** | **Time Occupied (s)** | **Gains (J)** |
| Extractor Fan | 42 | 0.66 | 2400 | 100800 |
| Hairdryer | 1600 | 0.166 | 600 | 960000 |
| | | | **Total (J)** | 1060800 |
| | | | **Total (W)** | **12.2778** |

**Total Appliance Gains Average over a one second snap shot of the day (W)**

**Total: 449.2278W**

The kitchen has the highest heat gain as they spend a lot of time there and there are a lot of appliances that are high in power. The room with the lowest heat gain is the bathroom as they spend the least amount of time in that room and it has only one high power appliance which is the extractor fan. The contrasting gains mean in summer the occupants might have to open a window in the kitchen and living room as it would get too hot.

## 6) Heating System Calculations

Converting heat losses from kWh to W:

| | Envelope | | | | | Ventilation | | | |
|---|---|---|---|---|---|---|---|---|---|
| | kWh | /hours per month | x1000 | W | | kWh | /hours per month | x1000 | W |
| Jan | 2681.6 | 3.604301075 | 3604.3 | 3604.3 | Jan | 1395.54 | 1.875725806 | 1875.73 | 3604.3 |
| Feb | 2665.83 | 3.967008929 | 3967.01 | 3967.01 | Feb | 1253.08 | 1.864702381 | 1864.7 | 3967.01 |
| Mar | 2360.57 | 3.17280914 | 3172.81 | 3172.81 | Mar | 1231.36 | 1.655053763 | 1655.05 | 3172.81 |
| Apr | 2018.03 | 2.802819444 | 2802.82 | 2802.82 | Apr | 1016.87 | 1.412319444 | 1412.32 | 2802.82 |
| May | 1545.87 | 2.077782258 | 2077.78 | 2077.78 | May | 804.49 | 1.081303763 | 1081.3 | 2077.78 |
| Sep | 1183.61 | 1.643902778 | 1643.9 | 1643.9 | Sep | 595.32 | 0.826833333 | 826.833 | 1643.9 |
| Oct | 1703.05 | 2.289045699 | 2289.05 | 2289.05 | Oct | 886.58 | 1.191639785 | 1191.64 | 2289.05 |
| Nov | 2255.81 | 3.133069444 | 3133.07 | 3133.07 | Nov | 1136.03 | 1.577819444 | 1577.82 | 3133.07 |
| Dec | 2605.54 | 3.502069892 | 3502.07 | 3502.07 | Dec | 1354.5 | 1.820560484 | 1820.56 | 3502.07 |

Heat losses VS gains table:

| | Heat Loss | | | Heat Gains (W) | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Envelope(W) | Ventilation (W) | Total(W) | Solar (W) | Lighting (W) | Metabolic (W) | Appliance (W) | Total (W) | Monthly Net Difference |
| Jan | 3604.30 | 3604.30 | **7208.60** | 111.15 | 154.58 | 236.23 | 449.23 | **951.19** | **6257.41** |
| Feb | 3967.01 | 3967.01 | **7934.02** | 205.40 | 154.58 | 236.23 | 449.23 | **1045.44** | **6888.58** |
| Mar | 3172.81 | 3172.81 | **6345.62** | 308.16 | 154.58 | 236.23 | 449.23 | **1148.20** | **5197.42** |
| Apr | 2802.82 | 2802.82 | **5605.64** | 433.44 | 154.58 | 236.23 | 449.23 | **1273.48** | **4332.16** |
| May | 2077.78 | 2077.78 | **4155.56** | 514.00 | 132.92 | 236.23 | 449.23 | **1332.38** | **2823.19** |
| Sep | 1643.90 | 1643.90 | **3287.81** | 359.63 | 112.08 | 236.23 | 449.23 | **1157.17** | **2130.64** |
| Oct | 2289.05 | 2289.05 | **4578.09** | 244.12 | 112.08 | 236.23 | 449.23 | **1041.66** | **3536.43** |
| Nov | 3133.07 | 3133.07 | **6266.14** | 136.29 | 142.08 | 236.23 | 449.23 | **963.83** | **5302.31** |
| Dec | 3502.07 | 3502.07 | **7004.14** | 92.97 | 142.08 | 236.23 | 449.23 | **920.51** | **6083.63** |
| | | | | | | | | Total (W) | **42551.7632** |

There is a net heat loss in all months except for the excluded months of June, July and August (cooling season). This is because of more heat coming in than escaping during these summer months. We do not need to heat these months, so I did not include them. There are great amounts of heat loss through the poor envelope construction with lack of thickness in insulation and also heat loss through ventilation of spaces.

| Monthly Net Difference | Number of seconds in a month (s) | Total Energy (J) | Energy consumption (kWh) |
|---|---|---|---|
| **6257.41** | 2678400.00 | 16759844641 | 4655.51 |
| **6888.58** | 2419200.00 | 16664853866 | 4629.13 |
| **5197.42** | 2678400.00 | 13920766778 | 3866.88 |
| **4332.16** | 2592000.00 | 11228952326 | 3119.15 |
| **2823.19** | 2678400.00 | 7561624672 | 2100.45 |
| **2130.64** | 2592000.00 | 5522618298 | 1534.06 |
| **3536.43** | 2678400.00 | 9471986966 | 2631.11 |
| **5302.31** | 2592000.00 | 13743583274 | 3817.66 |
| **6083.63** | 2678400.00 | 16294386536 | 4526.22 |
| | | **Total:** | **30880.17** |

**This means, to keep the house at 21 degrees Celsius, it requires 30880.17kWh every year.**

The efficiency of the heating system is 72%

| | Heating | Lighting and appliances |
|---|---|---|
| **Energy Consumed by the boiler (kWh)** | **C02 emissions (kg C02)** | **C02 emissions (kg C02)** |
| **(Energy needed by the house/0.72)** | **1kWh = 0.183kg C02** | **1kWh = 0.225kg C02** |
| 6465.99 | 1183.28 | 101.08 |
| 6429.34 | 1176.57 | 91.30 |
| 5370.67 | 982.83 | 101.08 |
| 4332.16 | 792.78 | 97.82 |
| 2917.29 | 533.86 | 97.45 |
| 2130.64 | 389.91 | 90.93 |
| 3654.32 | 668.74 | 93.96 |
| 5302.31 | 970.32 | 95.79 |
| 6286.41 | 1150.41 | 98.99 |
| **Total:** | **7848.71** | **868.39** |

Total Annual C02 emissions (kg C02 PA) is heating + electric = 7848.71+868.39 = **8717.10 kg C02 PA**

**'Normalise' these data according to total floor area (kWh/$m^2$ PA and $kgCO_2/m^2$ PA)**

| Building Element | Area (m2) | Energy Demand (kWh) | Per unit floor area (kWh/m2 PA) |
|---|---|---|---|
| Floor | 82.81 | 30880.17 | 372.91 |
| **Building Element** | **Area (m2)** | **Carbon emissions (kg C02)** | **Per unit floor area (kg C02/m2 PA)** |
| Floor | 82.81 | 8717.10 | 105.27 |

**Benchmarking:**

| Part L | Desired values | Current Values |
|---|---|---|
| Opening areas (window, roof window, roof lights and doors) | Not exceeding a total area of openings of 25% of the total floor area. | 26.80% |
| Walls | U = 0.18 W/(m^2K) | 0.576 W/(m^2K) |
| Floor | U = 0.13W/(m^2K) | 0.60 W/(m^2K) |
| Roof | U = 0.11 W/(m^2K) | 0.666 W/(m^2K) |
| Door | U = 1.0 W/(m^2K) | 2.5 W/(m^2K) |
| Window | U = 1.2W/(m^2K) | 2.8 W/(m^2K) |
| Air permeability | 5m^3/h m^2 at 50 Pa (0.249ach) | 1.1 W/(m^2K) |
| Boiler | Efficiency = 89.5% | 72% Efficiency |
| | | |
| **Part F** | | |
| Kitchen cooker hood | 30 l/s | 120 l/s |
| Bathroom extract fan | 15 l/s | 23 l/s |
| Whole dwelling ventilation rate | 0.3 l/s per m^2 (0.47 ach) | 1.1 ach |

Through this comparison, the house currently would not pass the most up to date Part L and F building regulations. As seen with red colouring, almost all values do not comply, only the kitchen cooker hood and bathroom extract fan do, marked in green.

**The overall performance of the house – comfort and satisfaction.**

In conclusion, poor construction causes significant heat loss. Ventilation is inadequate, with under-ventilation in the kitchen, over-ventilation from excessive infiltration, and an ineffective bathroom extract fan. While ample daylight improves thermal comfort via solar gains, the inefficient heating system increases energy waste and carbon emissions.